Experiment 23: Blood Management Information System

Aim:

To design and develop an interactive Blood Management Information System that maintains blood donor records, displays blood availability, and helps users manage blood donation information efficiently.

Algorithm:

1. Start the application and display the home page.
2. Provide options to view blood availability and register blood donors.
3. Collect donor details such as name, blood group, contact number, and location.
4. Validate the entered details and save donor records.
5. Allow users to search for donors using blood group and location filters.
6. Display available donor records and blood inventory information.
7. Provide an administrator dashboard to monitor donor records and donation statistics.
8. Display confirmation messages and stop the application when the user exits.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>BloodConnect | Blood Management System</title>
<style>
:root{--red:#dc2626;--dark:#171923;--muted:#64748b;--bg:#f6f7fb}
*{box-sizing:border-box}
body{margin:0;font-family:Arial,sans-serif;background:var(--bg);color:var(--dark)}
```

```
header{background:linear-gradient(120deg,#991b1b,#ef4444);color:white;padding:24px 6%;display:flex;align-items:center;justify-content:space-between;gap:15px;flex-wrap:wrap}

header h1{margin:0;font-size:25px}

header p{margin:7px 0 0;color:#fee2e2}

button{border:0;border-radius:9px;padding:11px 15px;font-weight:bold;cursor:pointer}

.primary{background:var(--red);color:white}

.light{background:white;color:#991b1b}

nav{display:flex;gap:8px;flex-wrap:wrap;padding:15px 6%;background:white;box-shadow:0 3px 12px #0000000b}

nav button{background:#f1f5f9;color:#334155}

nav button.active{background:#fee2e2;color:#991b1b}

main{max-width:1200px;margin:auto;padding:25px 5%}

.hero{background:white;border-radius:20px;padding:25px;display:grid;grid-template-columns:1.3fr 1fr;gap:20px;align-items:center;box-shadow:0 5px 25px #0f172a0c}

.hero h2{font-size:clamp(25px,4vw,38px);margin:0 0 12px}

.hero p{color:var(--muted);line-height:1.6}

.hero img{width:100%;height:220px;object-fit:cover;border-radius:15px}

.stats{display:grid;grid-template-columns:repeat(4,1fr);gap:14px;margin:20px 0}

.stat,.panel{background:white;padding:20px;border-radius:15px;box-shadow:0 5px 20px #0f172a08}

.stat span{color:var(--muted);font-size:13px}

.stat h2{margin:9px 0 0;color:var(--red);font-size:28px}

.panel{margin:16px 0}

.panel h3{margin-top:0}

input,select,textarea{width:100%;padding:12px;border:1px solid #dbe1ea;border-radius:9px;margin:6px 0 12px;font:inherit;background:white}

label{font-size:13px;font-weight:bold;color:#475569}

.grid{display:grid;grid-template-columns:repeat(2,minmax(0,1fr));gap:14px}

.row{display:flex;gap:10px;flex-wrap:wrap;align-items:center}
```

```
.row>*{flex:1}
table{width:100%;border-collapse:collapse;min-width:620px}
th,td{text-align:left;padding:12px;border-bottom:1px solid #e8edf3;font-size:14px}
th{background:#fff1f2;color:#991b1b}
.table-wrap{overflow-x:auto}
.tag{background:#fee2e2;color:#991b1b;padding:5px 9px;border-radius:20px;font-weight:bold}
.hidden{display:none!important}
.muted{color:var(--muted);font-size:13px}
.empty{text-align:center;padding:25px;color:var(--muted)}
footer{text-align:center;padding:25px;color:var(--muted);font-size:13px}
@media(max-width:750px){.hero{grid-template-columns:1fr}.stats{grid-template-columns:repeat(2,1fr)}.grid{grid-template-columns:1fr}main{padding:18px 4%}}
</style>
</head>
<body>
<header>
 <div>
  <h1>🩸 BloodConnect</h1>
  <p>Blood Management Information System</p>
 </div>
 <button class="light" onclick="showPage('register')">＋ Become a Donor</button>
</header>

<nav>
 <button class="active" data-page="home" onclick="showPage('home',this)">Dashboard</button>
 <button data-page="search" onclick="showPage('search',this)">Find Donors</button>
```

```
 <button data-page="register" onclick="showPage('register',this)">Register Donor</button>
 <button data-page="donations" onclick="showPage('donations',this)">Donation Records</button>
 <button data-page="admin" onclick="showPage('admin',this)">Admin</button>
</nav>

<main>
<section id="home" class="page">
 <div class="hero">
  <div>
   <span class="tag">Every drop matters</span>
   <h2>Give blood.<br>Help save lives.</h2>
   <p>Maintain donor information, search by blood group and location, and monitor donation records in one simple dashboard.</p>
   <button class="primary" onclick="showPage('search')">Find a Donor →</button>
   <button onclick="showPage('register')">Register Now</button>
  </div>
  <img src="https://images.unsplash.com/photo-1615461066841-6116e61058f4?auto=format&fit=crop&w=1000&q=80" alt="Blood donation">
 </div>
 <div class="stats">
  <div class="stat"><span>Registered Donors</span><h2 id="totalDonors">0</h2></div>
  <div class="stat"><span>Donation Records</span><h2 id="totalDonations">0</h2></div>
  <div class="stat"><span>Blood Groups</span><h2 id="bloodGroups">0</h2></div>
  <div class="stat"><span>Locations Covered</span><h2 id="locations">0</h2></div>
 </div>
 <div class="panel">
```

```
  <h3>Blood Group Overview</h3>
  <div id="bloodOverview" class="grid"></div>
  <p class="muted">Counts represent registered donors, not verified blood-bank inventory.</p>
 </div>
</section>

<section id="search" class="page hidden">
 <div class="panel">
  <h2>Find a Blood Donor</h2>
  <p class="muted">Search the locally stored donor directory by blood group or location.</p>
  <div class="grid">
   <div><label>Blood Group</label>
    <select id="filterGroup" onchange="renderDonors()">
     <option value="">All blood groups</option>
     <option>A+</option><option>A-</option><option>B+</option><option>B-</option>
     <option>AB+</option><option>AB-</option><option>O+</option><option>O-</option>
    </select>
   </div>
   <div><label>Location</label><input id="filterLocation" placeholder="Enter city or area" oninput="renderDonors()"></div>
  </div>
  <div class="table-wrap"><table>
   <thead><tr><th>Name</th><th>Blood Group</th><th>Location</th><th>Contact</th></tr></thead>
   <tbody id="donorResults"></tbody>
  </table></div>
```

```
  <p class="muted">Demo directory only. Verify donor eligibility and availability through authorized services before arranging a donation.</p>
 </div>
</section>

<section id="register" class="page hidden">
 <div class="panel">
  <h2>Donor Registration</h2>
  <form id="donorForm">
   <div class="grid">
    <div><label>Full Name</label><input name="name" required maxlength="70" placeholder="Enter full name"></div>
    <div><label>Blood Group</label>
     <select name="group" required>
      <option value="">Select group</option>
      <option>A+</option><option>A-</option><option>B+</option><option>B-</option>
      <option>AB+</option><option>AB-</option><option>O+</option><option>O-</option>
     </select>
    </div>
    <div><label>Contact Number</label><input name="phone" required pattern="[0-9+ -]{8,16}" placeholder="Phone number"></div>
    <div><label>City / Location</label><input name="location" required maxlength="80" placeholder="City or area"></div>
    <div><label>Age</label><input name="age" type="number" min="18" max="100" required></div>
    <div><label>Last Donation Date (optional)</label><input name="lastDonation" type="date"></div>
   </div>
```

```
    <label><input type="checkbox" required style="width:auto"> I confirm these details are accurate and consent to this demo record being stored in this browser.</label>
    <br><br><button class="primary" type="submit">Save Donor Record</button>
   </form>
  </div>
</section>

<section id="donations" class="page hidden">
  <div class="panel">
   <h2>Donation Records</h2>
   <p class="muted">Record a donation only after confirming it through the relevant blood bank or authorized staff.</p>
   <form id="donationForm">
    <div class="grid">
     <div><label>Donor</label><select id="donorSelect" required></select></div>
     <div><label>Donation Date</label><input id="donationDate" type="date" required></div>
    </div>
    <button class="primary">Add Donation Record</button>
   </form>
   <div class="table-wrap"><table>
    <thead><tr><th>Record ID</th><th>Donor</th><th>Blood Group</th><th>Date</th></tr></thead>
    <tbody id="donationResults"></tbody>
   </table></div>
  </div>
</section>

<section id="admin" class="page hidden">
```

```
 <div class="panel" id="adminLogin">
  <h2>Administrator Login</h2>
  <p class="muted">For classroom demonstration only. Default demo password: <b>admin123</b></p>
  <label>Password</label><input id="adminPassword" type="password" placeholder="Enter demo password">
  <button class="primary" onclick="loginAdmin()">Sign In</button>
  <p id="loginMessage" class="muted"></p>
 </div>
 <div id="adminPanel" class="panel hidden">
  <div class="row"><h2>Admin Dashboard</h2><button onclick="exportCSV()">Export Donors CSV</button></div>
  <p class="muted">Manage the demo donor directory. Do not enter real sensitive medical or contact information.</p>
  <div class="table-wrap"><table>

<thead><tr><th>Name</th><th>Group</th><th>Location</th><th>Action</th></tr></thead>
    <tbody id="adminResults"></tbody>
  </table></div>
  <button onclick="logoutAdmin()">Logout</button>
 </div>
</section>
</main>

<footer>BloodConnect • Internet Programming Lab • Educational demo — not a real blood bank</footer>

<script>
const KEY = "bloodconnect_donors_v1";
```

```
const DONATIONS_KEY = "bloodconnect_donations_v1";
const groups = ["A+","A-","B+","B-","AB+","AB-","O+","O-"];

function read(key) {
 try { return JSON.parse(localStorage.getItem(key) || "[]"); }
 catch { return []; }
}
function save(key, data) { localStorage.setItem(key, JSON.stringify(data)); }

let donors = read(KEY);
let donations = read(DONATIONS_KEY);
let adminLoggedIn = false;

if (!donors.length) {
 donors = [
  {id:"D1001",name:"Arun
Kumar",group:"O+",phone:"9000000001",location:"Chennai",age:25,lastDonation:""},
  {id:"D1002",name:"Priya
S",group:"A+",phone:"9000000002",location:"Coimbatore",age:23,lastDonation:""},
  {id:"D1003",name:"Rahul
M",group:"B+",phone:"9000000003",location:"Chennai",age:28,lastDonation:""},
  {id:"D1004",name:"Meena
R",group:"AB+",phone:"9000000004",location:"Madurai",age:24,lastDonation:""}
 ];
 save(KEY, donors);
}

function escapeHTML(value) {
 return String(value ?? "").replace(/[&<>"']/g, c =>
```

```
  ({"&":"&amp;","<":"&lt;",">":"&gt;","'":"&quot;",'"':"&#39;"}[c]));
}

function showPage(id, button) {
 document.querySelectorAll(".page").forEach(p => p.classList.add("hidden"));
 document.getElementById(id).classList.remove("hidden");
 document.querySelectorAll("nav button").forEach(b => b.classList.remove("active"));
 const navButton = button || document.querySelector('nav button[data-page="'+id+'"]');
 if (navButton) navButton.classList.add("active");
 refresh();
}

document.getElementById("donorForm").addEventListener("submit", e => {
 e.preventDefault();
 const f = new FormData(e.target);
 const donor = {
  id:"D" + Date.now().toString().slice(-8),
  name:f.get("name").trim(),
  group:f.get("group"),
  phone:f.get("phone").trim(),
  location:f.get("location").trim(),
  age:Number(f.get("age")),
  lastDonation:f.get("lastDonation") || ""
 };
 if (!donor.name || !donor.location || !donor.group) return;
 donors.push(donor);
 save(KEY, donors);
 e.target.reset();
```

```
  alert("Donor record saved! Reference ID: " + donor.id);
  refresh();
  showPage("search");
});

function renderDonors() {
  const group = document.getElementById("filterGroup").value;
  const location =
document.getElementById("filterLocation").value.trim().toLowerCase();
  const found = donors.filter(d =>
    (!group || d.group === group) &&
    (!location || d.location.toLowerCase().includes(location))
  );
  document.getElementById("donorResults").innerHTML = found.length
    ? found.map(d => `<tr><td>${escapeHTML(d.name)}</td><td><span
class="tag">${escapeHTML(d.group)}</span></td><td>${escapeHTML(d.location)}</td
><td>${escapeHTML(d.phone)}</td></tr>`).join("")
    : '<tr><td colspan="4" class="empty">No matching donors found.</td></tr>';
}

function renderOverview() {
  document.getElementById("bloodOverview").innerHTML = groups.map(g => {
    const count = donors.filter(d => d.group === g).length;
    return `<div class="stat"><span>${g} Donors</span><h2>${count}</h2></div>`;
  }).join("");
}

function renderDonations() {
  document.getElementById("donorSelect").innerHTML = donors.length
```

```
    ? donors.map(d => `<option value="${escapeHTML(d.id)}">${escapeHTML(d.name)} (${escapeHTML(d.group)})</option>`).join("")
    : '<option value="">Register a donor first</option>';

  document.getElementById("donationResults").innerHTML = donations.length
    ? donations.slice().reverse().map(x => {
      const d = donors.find(y => y.id === x.donorId);
      return `<tr><td>${escapeHTML(x.id)}</td><td>${escapeHTML(d ? d.name : "Removed donor")}</td><td>${escapeHTML(d ? d.group : "—")}</td><td>${escapeHTML(x.date)}</td></tr>`;
    }).join("")
    : '<tr><td colspan="4" class="empty">No donation records yet.</td></tr>';
}

document.getElementById("donationForm").addEventListener("submit", e => {
  e.preventDefault();
  const donorId = document.getElementById("donorSelect").value;
  const date = document.getElementById("donationDate").value;
  if (!donors.some(d => d.id === donorId) || !date) {
    alert("Please register/select a donor and choose a date.");
    return;
  }
  donations.push({id:"BR" + Date.now().toString().slice(-7), donorId, date});
  save(DONATIONS_KEY, donations);
  e.target.reset();
  refresh();
  alert("Donation record added.");
});
```

```
function loginAdmin() {
  if (document.getElementById("adminPassword").value === "admin123") {
    adminLoggedIn = true;
    document.getElementById("adminLogin").classList.add("hidden");
    document.getElementById("adminPanel").classList.remove("hidden");
    renderAdmin();
  } else {
    document.getElementById("loginMessage").textContent = "Incorrect password.";
  }
}

function logoutAdmin() {
  adminLoggedIn = false;
  document.getElementById("adminLogin").classList.remove("hidden");
  document.getElementById("adminPanel").classList.add("hidden");
  document.getElementById("adminPassword").value = "";
}

function renderAdmin() {
  if (!adminLoggedIn) return;
  document.getElementById("adminResults").innerHTML = donors.map(d =>

`<tr><td>${escapeHTML(d.name)}</td><td>${escapeHTML(d.group)}</td><td>${escapeHTML(d.location)}</td><td><button onclick="deleteDonor('${escapeHTML(d.id)}')">Delete</button></td></tr>`

  ).join("") || '<tr><td colspan="4">No donor records.</td></tr>';
}

function deleteDonor(id) {
```

```
  if (!adminLoggedIn || !confirm("Delete this donor record?")) return;
  donors = donors.filter(d => d.id !== id);
  save(KEY, donors);
  refresh();
}

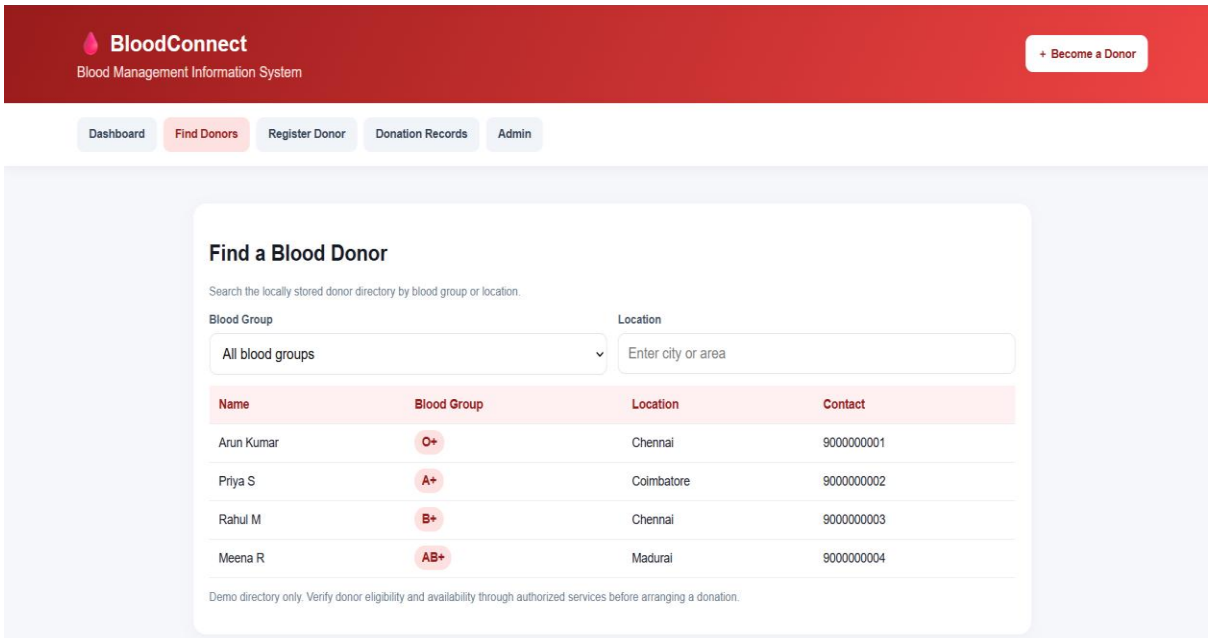
function exportCSV() {
  if (!adminLoggedIn) return;
  const rows = [["ID","Name","Blood Group","Phone","Location","Age"],
    ...donors.map(d => [d.id,d.name,d.group,d.phone,d.location,d.age])];
  const csv = rows.map(row => row.map(v =>
    '"' + String(v ?? "").replace(/"/g,'""') + '"'
  ).join(",")).join("\r\n");
  const blob = new Blob([csv], {type:"text/csv;charset=utf-8;"});
  const url = URL.createObjectURL(blob);
  const a = document.createElement("a");
  a.href = url; a.download = "bloodconnect-donors.csv"; a.click();
  URL.revokeObjectURL(url);
}

function refresh() {
  donors = read(KEY);
  donations = read(DONATIONS_KEY);
  document.getElementById("totalDonors").textContent = donors.length;
  document.getElementById("totalDonations").textContent = donations.length;
  document.getElementById("bloodGroups").textContent = new Set(donors.map(d =>
d.group)).size;
  document.getElementById("locations").textContent = new Set(donors.map(d =>
d.location.toLowerCase())).size;
```

```
  renderOverview();
  renderDonors();
  renderDonations();
  renderAdmin();
}

window.addEventListener("storage", refresh);
refresh();
</script>
</body>
</html>
```

Output:



Result:

Thus, the Blood Management Information System was developed using HTML, CSS, and JavaScript. It supports donor registration, blood-group-based searching, donation record management, and administrative monitoring through browser storage.